

Automation and programming

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WHY? We want to ...

1. ... avoid tedious manual labor (lazy)
2. ... make sure that we work correctly
3. ... be efficient: work faster, with less energy
4. ... make our work reproducible

which applies both to...

- managing information, files and documents
- calculations, analyzing data



Instructions?

- Statements that follow each other
- Every statement does something to change the state of the computer
- Linear sequence
- How can a computer understand what we are telling it?
- Multiple levels, exact instructions combine them

Pancake Recipe

- 100g plain flour
- 2 eggs
- 300ml milk
- 1 tbsp oil
- pinch of salt



1. Put the flour and milk into a bowl.
2. Crack the eggs and add to the bowl.
3. Whisk the ingredients together.
4. Pour some of the mixture into the pan.
5. Cook until browned then flip.
6. Once the other side is brown leave to cool.
7. Enjoy eating.

Programming, again...

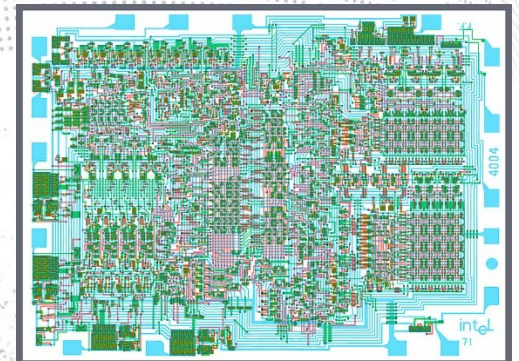
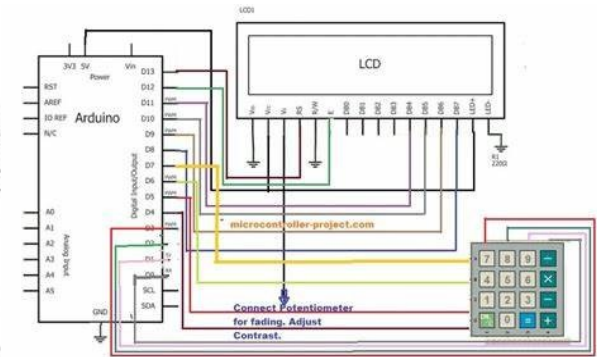


- The concept of calculation: how much is $651/7$?

You have 651 balls.

1. You go through them one-by-one.
2. You put every 7th ball in a bin.
3. After done, count the balls. (divisor)

- You can do this with electricity
- You are using a machine to define a machine that calculates numbers that represent something else (programmable computer)



What kind of languages are there?

General purpose vs. **specialized** (e.g. domain-specific) language



What kind of languages are there?

Different **levels** of programming

MONITOR FOR 6802 1.4 9-14-80 TSC ASSEMBLER PAGE 2

```
C000          ORG      RCM+$0000 BEGIN MONITOR
C000 8E 00 70  START  LDS      #STACK
```

```
*****
* FUNCTION: INITA - Initialize ACIA
* INPUT: none
* OUTPUT: none
* CALLS: none
* DESTROYS: acc A
```

```
0013      RESETA EQU    $00010011
0011      CTLREG EQU    $00010001
```

```
C003 86 13      INITA  LDA A  #RESETA  RESET ACIA
C005 B7 80 04      STA A  ACIA
C008 86 11      LDA A  #CTLREG  SET 8 BITS AND 2 STOP
C00A B7 80 04      STA A  ACIA
C00D 7E C0 F1      JMP      SIGNON  GO TO START OF MONITOR
```

```
*****
* FUNCTION: INCH - Input character
* INPUT: none
* OUTPUT: char in acc A
* DESTROYS: acc A
* CALLS: none
* DESCRIPTION: Gets 1 character from terminal
```

```
C010 B6 80 04  INCH  LDA A  ACIA      GET STATUS
C013 47          ASR A          SHIFT RDRF FLAG INTO CARRY
C014 24 FA      BCC  INCH      RECIEVE NOT READY
C016 B6 80 05      LDA A  ACIA+1  GET CHAR
C019 84 7F      AND A  #$7F      MASK PARITY
C01B 7E C0 79      JMP      OUTCH  ECHO & RTS
```

```
*****
* FUNCTION: INHEX - INPUT HEX DIGIT
* INPUT: none
* OUTPUT: Digit in acc A
* CALLS: INCH
* DESTROYS: acc A
* Returns to monitor if not HEX input
```

```
C01E 8D F0  INHEX  BSR  INCH  GET A CHAR
C020 81 30      CMP A  #'0      ZERO
C022 2B 11      BMI  HEXERR  NOT HEX
C024 81 39      CMP A  #'9      NINE
C026 2F 0A      BLE  HEXRTS  GOOD HEX
C028 81 41      CMP A  #'A      NOT HEX
C02A 2B 09      BMI  HEXERR
C02C 81 46      CMP A  #'F
C02E 2E 05      BGT  HEXERR
C030 80 07      SUB A  #7      FIX A-F
C032 84 0F  HEXRTS AND A  #$0F  CONVERT ASCII TO DIGIT
C034 39      RTS
```

```
C035 7E C0 AF  HEXERR JMP      CTRL  RETURN TO CONTROL LOOP
```

```
INPUT CELSIUS_TEMP
SET FAHRENHEIT_TEMP TO CELSIUS_TEMP * 9/5 + 32
WRITE FAHRENHEIT_TEMP
```



```
makeEggs()
getEggs()
stirEggs()
cook()
serve()
```

High-Level Languages
(Java, PHP, Python, etc.)

Assembly
Language

Assembler

Machine Code

Instruction
set

Hardware

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Low level program



1 second



1 year



High level program



5 seconds



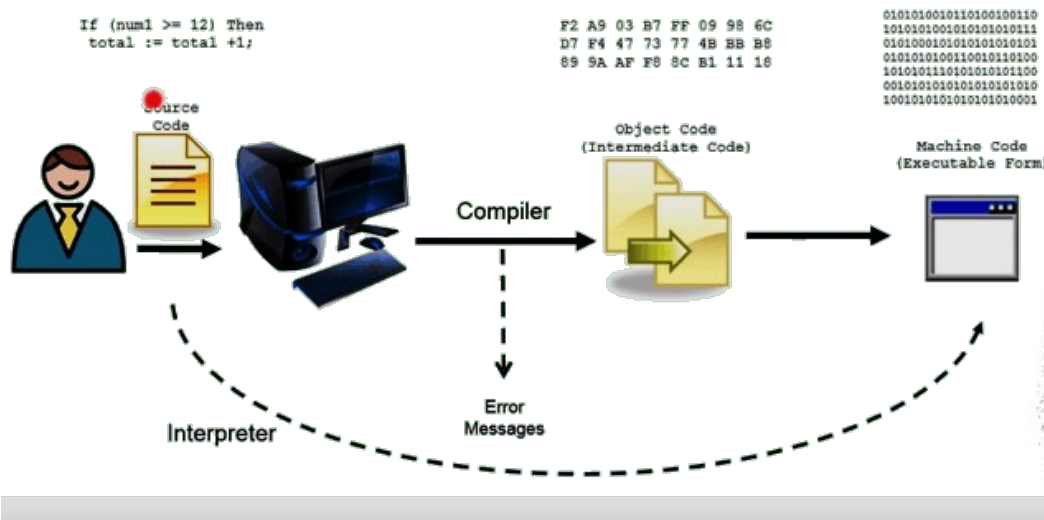
1 month



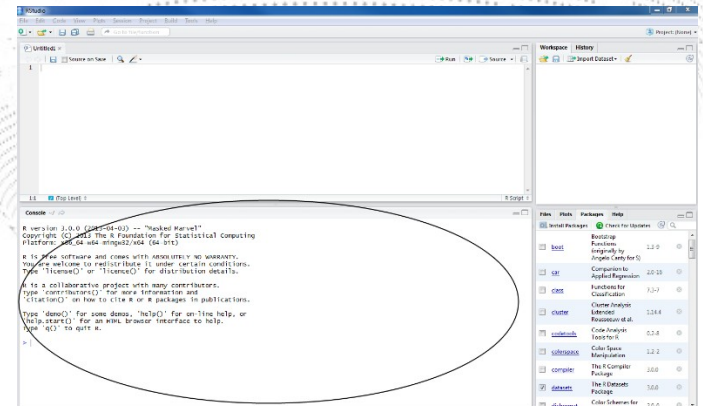
What kind of languages are there?

Interpreted vs. compiled languages

Compilers & Interpreters (high-level)



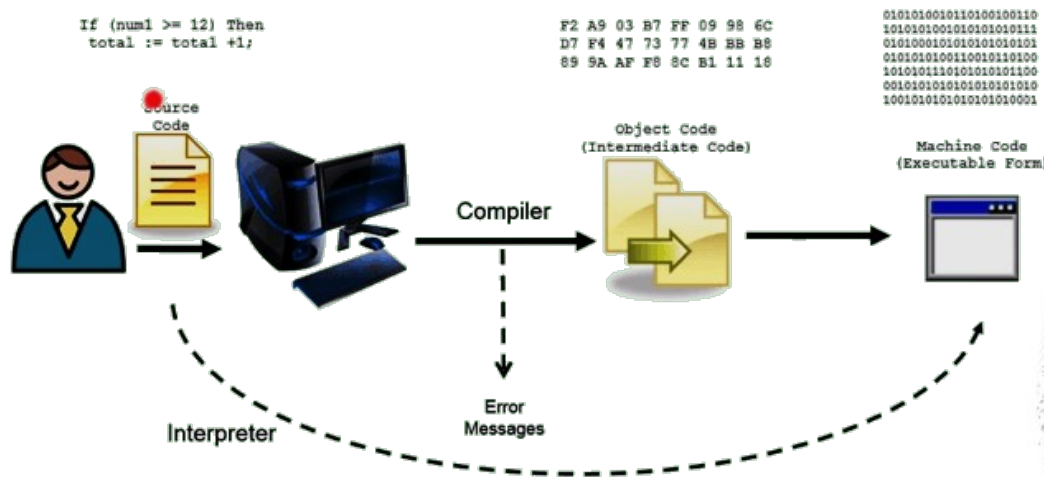
Working in a console



What kind of languages are there?

Interpreted vs. **compiled** languages

Compilers & Interpreters (high-level)



```
1 7f45 4c46 0201 0100 0000 0000 0000 0000
2 0300 3e00 0100 0000 b007 0000 0000 0000
3 4000 0000 0000 0000 981b 0000 0000 0000
4 0000 0000 4000 3800 0900 4000 1d00 1c00
5 0600 0000 0400 0000 4000 0000 0000 0000
6 4000 0000 0000 0000 4000 0000 0000 0000
7 f801 0000 0000 0000 f801 0000 0000 0000
8 0800 0000 0000 0000 0300 0000 0400 0000
9 3802 0000 0000 0000 3802 0000 0000 0000
10 3802 0000 0000 0000 1c00 0000 0000 0000
11 1c00 0000 0000 0000 0100 0000 0000 0000
12 0100 0000 0500 0000 0000 0000 0000 0000
13 0000 0000 0000 0000 0000 0000 0000 0000
14 780b 0000 0000 0000 780b 0000 0000 0000
15 0000 2000 0000 0000 0100 0000 0600 0000
16 780d 0000 0000 0000 780d 2000 0000 0000
17 780d 2000 0000 0000 9802 0000 0000 0000
18 c003 0000 0000 0000 0000 2000 0000 0000
19 0200 0000 0600 0000 900d 0000 0000 0000
20 900d 2000 0000 0000 900d 2000 0000 0000
21 0002 0000 0000 0000 0002 0000 0000 0000
22 0800 0000 0000 0000 0400 0000 0400 0000
23 5402 0000 0000 0000 5402 0000 0000 0000
24 5402 0000 0000 0000 4400 0000 0000 0000
25 4400 0000 0000 0000 0400 0000 0000 0000
26 50e5 7464 0400 0000 e409 0000 0000 0000
```

```
adam@vulcanodon:~$ chmod +x hello_exec
adam@vulcanodon:~$ ./hello_exec
Hello World!
adam@vulcanodon:~$
```

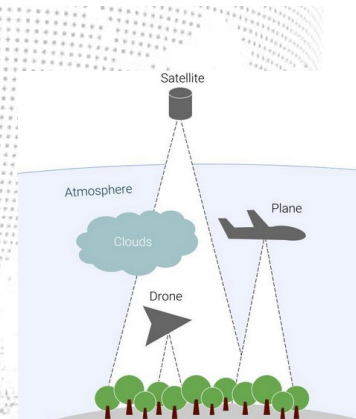
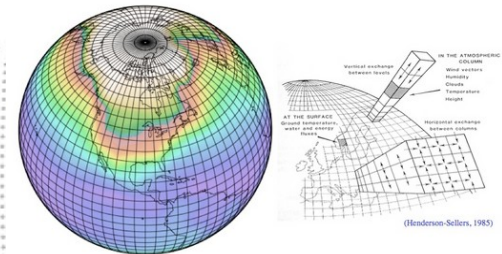
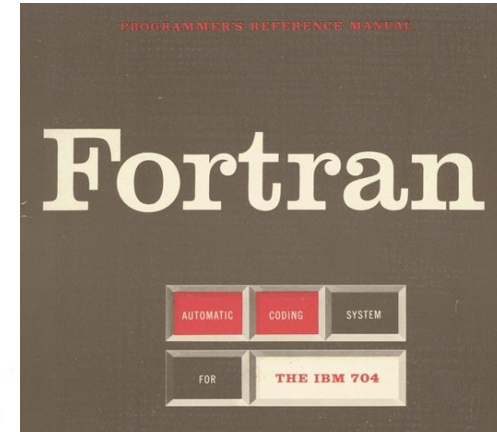
```
adam@vulcanodon:~$ g++ hello.cpp -o hello_exec
```

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     cout << "Hello World!\n";
7     return 0;
8 }
9
```

Some programming languages...

Fortran

- IBM, since 1957, first high-level language
- For mathematical computations, compiled
- One of the fastest languages, still
- climate models, remote sensing data, crystallography

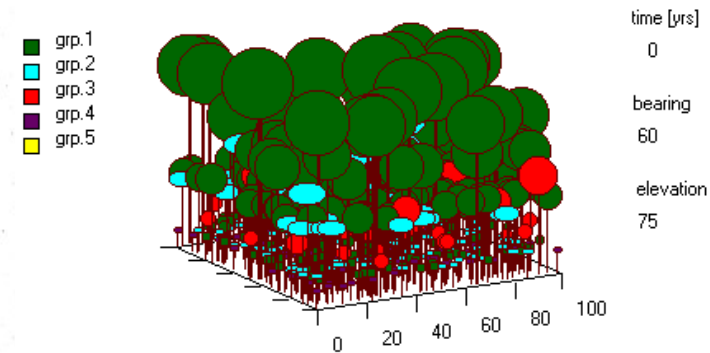


Some programming languages... C++

- C with extended object-oriented features
- Complex data structures, yet very fast
- Used everywhere, popular desktop applications (e.g. Adobe PS, MS Office) computer games, agent-based modelling
- Very good R integration (Rcpp package)



C++ is the new C — twice the power, twice the size, works in hostile environments, and if you try to use it without care and special training you will probably crash.



Some programming languages... Java

- Based on C too
- Compiled, runs in a virtual machine: code is very deployable
- Faster than either R or Python
- Some desktop applications, mesquite, imageJ



Java is another attempt to improve on C. It sort of gets the job done, but it's way slower, bulkier, spews pollution everywhere, and people will think you're a redneck.



Some programming languages

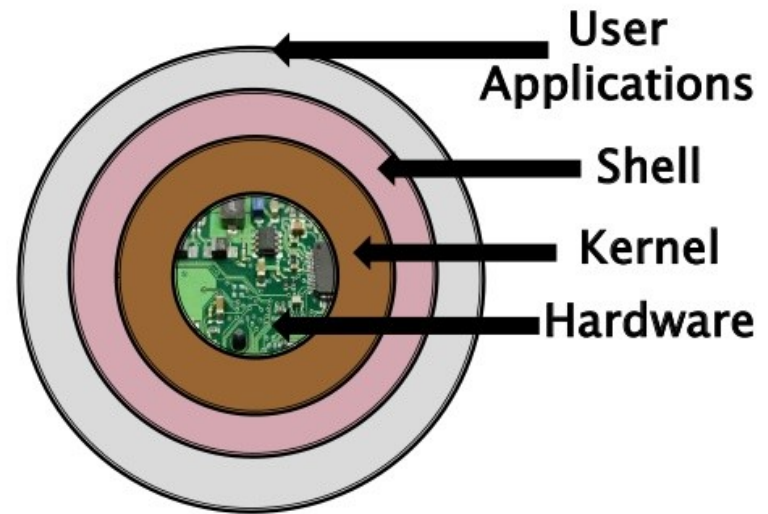
JavaScript

- Scripting language for the World Wide Web
- Executed by the clients (the computer visiting the website)
- Controls animations, interactive content
- Application development using node.js



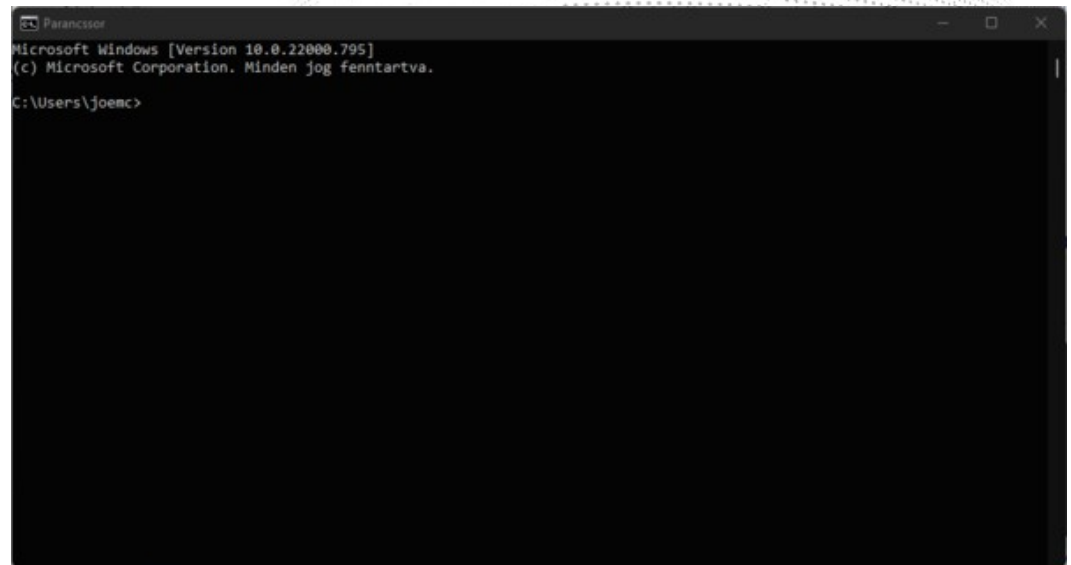
Some programming languages... **bash**

- Shell scripting language based on the Bourne shell (1976)
- Current standard on unix-like operating systems (e.g. Linux)
- Useful for file management, system administration running console applications, raw data processing
- Other solutions: zsh, ksh, csh, fish



Some programming languages... cmd.exe

- The command prompt
- The shell of Windows
- Very tedious to use

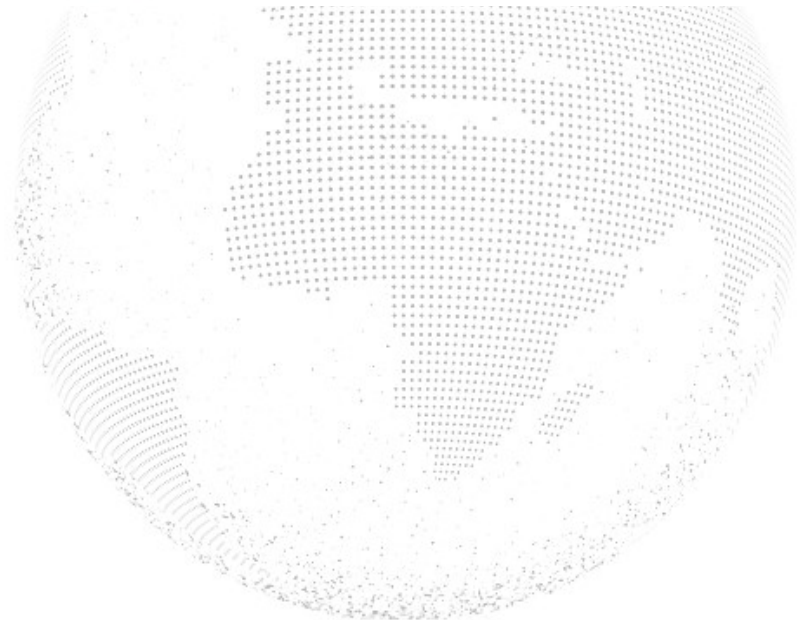


Some programming languages... Python

- Since 1991
- Higher level than C, interpreted, general purpose
- Very popular due to the clean syntax
- Two main version still in use: Python 2 and Python 3
- Tons of scientific packages, many programs have python APIs
- Desktop applications, e.g. Gplates, debian-apt



Python is great for everyday tasks: easy to drive, versatile, comes with all the conveniences built in. It isn't fast or sexy, but neither are your errands.

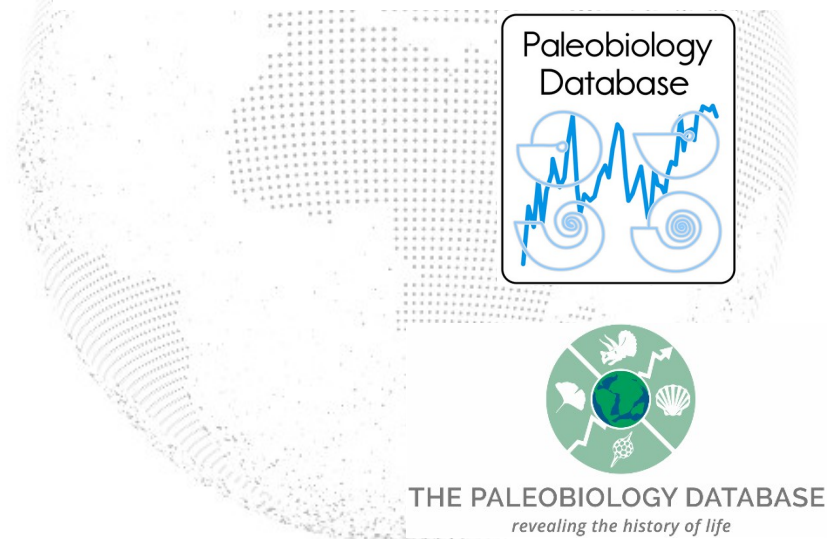


Some programming languages... Perl

- A family of languages
- Originally for text processing, somewhat faster than python
- Used commonly in bioinformatics, e.g. DNA sequence analysis
- Sometimes for the web with databases (originally the PaleoDB website was using perl)



Perl used to serve the same purpose as Python, but now only bearded ex-hippies use it.

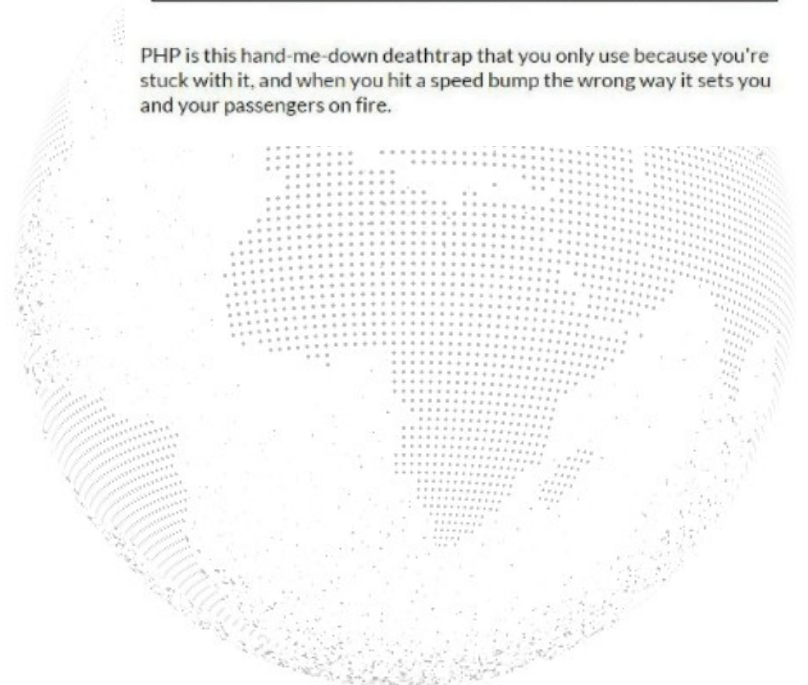


Some programming languages... **PHP**

- General purpose, designed for web development, interpreted
- Server-side programming
- Very good database integration
- Web-based applications, shops, content management (e.g. Wordpress)



PHP is this hand-me-down deathtrap that you only use because you're stuck with it, and when you hit a speed bump the wrong way it sets you and your passengers on fire.



Some programming languages... SQL

- Structured Query Language
- The language of relational databases
- Local databases: MySQL, PostgreSQL, MariaDB, Oracle Database
- Define, Manage, Query



Some programming languages... **MATLAB**

- Since 1984, Mathworks
- Mathematical computations, especially linear algebra
- Proprietary – good packages
- GNU alternative: GNU Octave
- Many mathematical, engineering, scientific algorithms are only available in this language

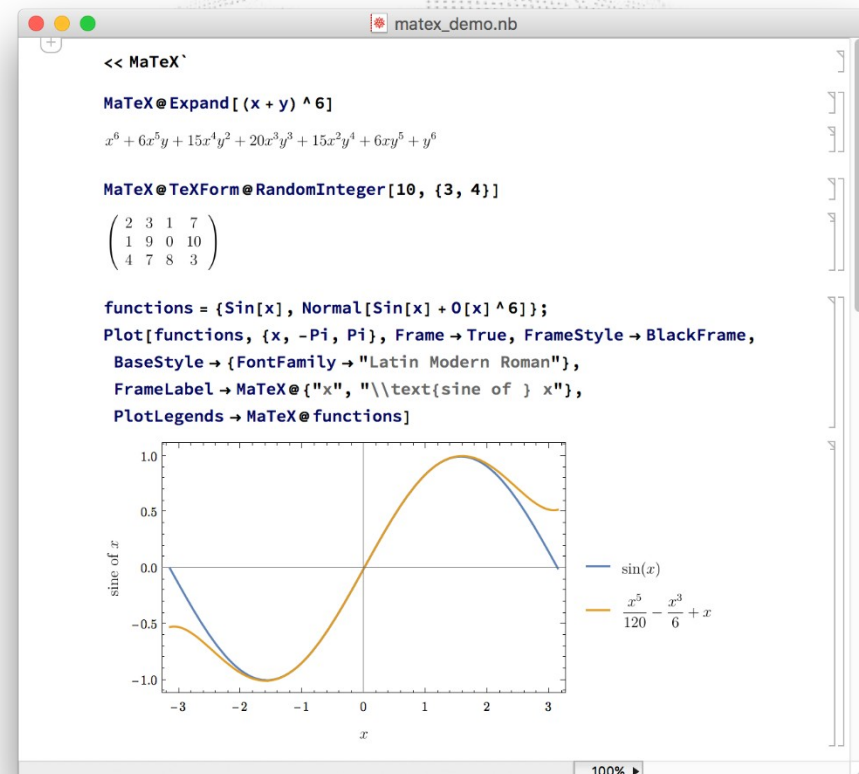


MATLAB is what scientists use to do special scientist things.



Some programming languages... Mathematica

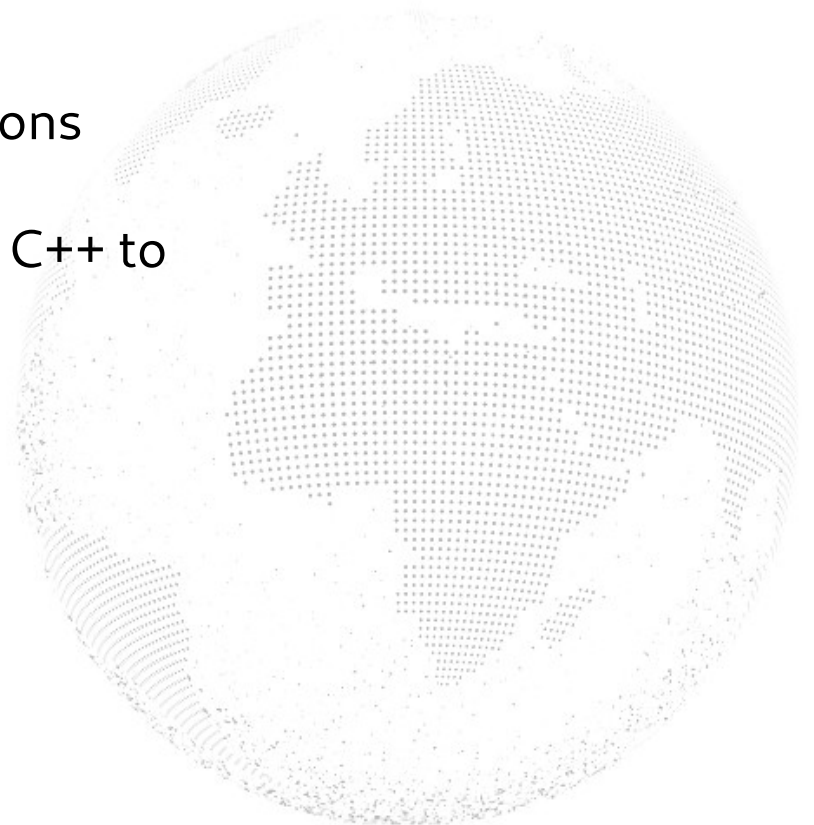
- Developed by Wolfram Research
- Symbolic language, as close to maths as possible
- Alternative to matlab (even more expensive)



Some more... Julia



- The next big thing, growing fast
- Built for ease of use and performance at the same time
- Good choice for numerical simulations
- Only language besides Fortran and C++ to reach petaflops-level performance



Some even more...

- BASIC, C#, Rust, Lua, Ruby, Go, Pascal, LISP, F#, Haskell, SAS, etc..



```
READY
10 FOR X=1 TO 10
20 PRINT "HELLO WIKIPEDIA"
30 NEXT X
RUN
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
HELLO WIKIPEDIA
READY
■
```



https://en.wikipedia.org/wiki/List_of_programming_languages

R (S)

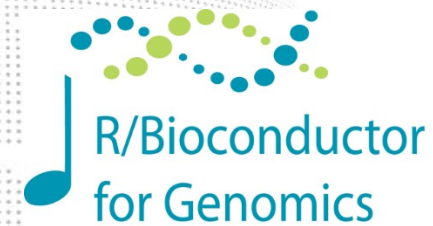
- GNU version of **S** (1976), since 1992 (**R**oss Ihaka and **R**obert Gentleman, cf. S3, S4)
- Written mostly in C and Fortran
- Statistics-oriented
- 9th most popular language on TIOBE
- High-level language: can be very slow
- Interpreted
- **CRAN** packages (25355)
- Contributor to Debian



R is what scientists use when they can't afford MATLAB.



Package	Package	Package
base	"boot"	"recommended"
base	"base"	"base"
base	"boot"	"recommended"
base	"class"	"recommended"
cluster	"cluster"	"recommended"
codetools	"codetools"	"recommended"
compiler	"compiler"	"base"
datasets	"datasets"	"base"
foreign	"foreign"	"recommended"
graphics	"graphics"	"base"
grDevices	"grDevices"	"base"
grid	"grid"	"base"
KernSmooth	"KernSmooth"	"recommended"
lattice	"lattice"	"recommended"
MASS	"MASS"	"recommended"
Matrix	"Matrix"	"recommended"
methods	"methods"	"base"
mgcv	"mgcv"	"recommended"
nme	"nme"	"recommended"
nnet	"nnet"	"recommended"
parallel	"parallel"	"base"
rpart	"rpart"	"recommended"
spatial	"spatial"	"recommended"
splines	"splines"	"base"
stats	"stats"	"base"
stats4	"stats4"	"base"
survival	"survival"	"recommended"
tcltk	"tcltk"	"base"
tools	"tools"	"base"
utils	"utils"	"base"



R is not RStudio!



R: Language, tools to use it

- Terminal
- Plotting 'devices'



Rstudio: Integrated Development Environment (IDE) for R

- Runs R
- Code editor
- Document Building

